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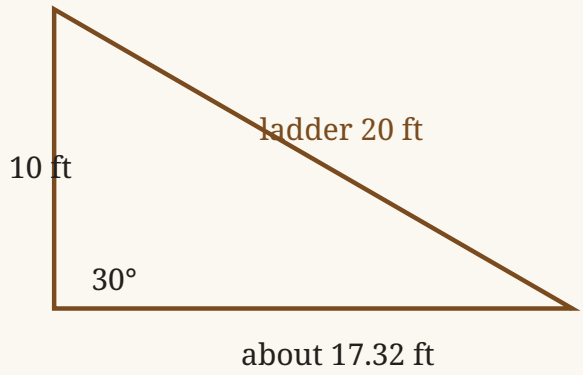
One ladder: sine, cosine, and tangent

The numbers are a paper example. They are not instructions for setting a real ladder.

Picture a ladder against a straight wall on level ground. The ladder, the wall, and the ground make a triangle. The same triangle is used throughout.

Trigonometry connects the angles of a triangle with the lengths of its sides. This page does not need calculus.

Same triangle, three questions



Sine: height / ladder

Cosine: base / ladder

Tangent: height / base

Paper example. Not a rule for setting a real ladder.

A right triangle: 20-foot ladder, 10-foot height, base about 17.32 feet, 30 degrees at the ground

Paper triangle. The square corner is where the wall meets the ground.

A right triangle has one square corner, 90 degrees. Here that corner is at the foot of the wall. The hypotenuse, pronounced hy-POT-uh-noos, is the side across from the square corner. It is the longest side. Here it is the ladder, 20 feet. An angle measures how wide two lines open. The ground angle, from the ground up to the ladder, is 30 degrees. The sign ° means degrees.

Relative to that ground angle, the height is the opposite side: it sits across from the angle. The base is the adjacent side: it touches the angle and is not the hypotenuse.

Sine: how high

Sine, pronounced like "sign," is the height divided by the ladder length. A ratio is one length compared with another by division. It is not itself a height.

For 30 degrees, the height is half the ladder. A calculator writes sin(30°) = 0.5. The short name sin means sine. So the height is 20 feet times 0.5, which is 10 feet. The sine is the ratio 0.5. Multiplying by the ladder turns the ratio into a length.

That half is not a mysterious calculator fact. An equilateral triangle has three equal sides and three 60-degree angles. Split one with a 20-foot side down the middle and you get two right triangles. The 60-degree top angle becomes two 30-degree angles, and the base becomes two 10-foot pieces. The side opposite 30 degrees is 10 feet. The hypotenuse is still 20. Ten divided by 20 is one half. Turn that half-triangle over and it is the ladder.

Cosine: how far out

Cosine, pronounced "co-sign," is the base divided by the ladder. For 30 degrees a calculator gives about 0.866. The short name is cos. The base is 20 times 0.866, about 17.32 feet. The sign ≈ means approximately equal. The 0.866 is rounded.

Tangent: height from the base

Tangent, shortened to tan, is the height divided by the base. If you know the ground angle and the distance from the wall, and you do not know the ladder, tangent still reaches the height. For 30 degrees, tan is about 0.577. Every foot along the ground goes with about 0.577 feet of rise. So 17.32 times 0.577 is about 10 feet again.

The ratios stay put when the triangle grows and the angle does not. A 40-foot ladder at 30 degrees reaches 20 feet. Double the size, keep the shape. 10 divided by 20 equals 20 divided by 40, and both equal 0.5. Change the angle and the ratios change.

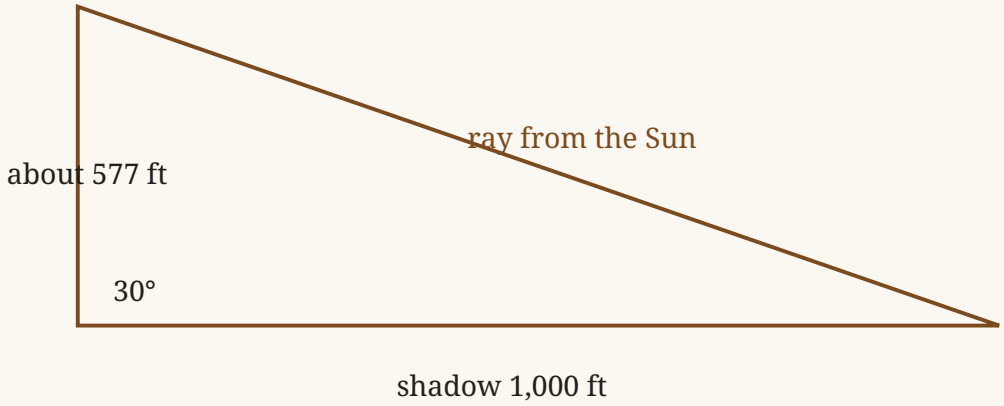
On a calculator, choose degrees, marked DEG, not radians. A radian is another unit for the same opening. sin, cos, and tan then take 30 as 30 degrees.

From a ladder to a lunar landscape

A shadow makes the same triangle. Replace the wall height with a mountain's height. Replace the base with its shadow on level ground. A ray of sunlight takes the place of the sloping ladder.

The Sun's elevation angle means how high the Sun is above the local horizon, the level direction at that place. Here the angle is 30 degrees again, so the same tangent applies.

Invented example, not Galileo's data



The upright line is the mountain's height. Level ground and one sun angle are assumed.

A right triangle: mountain height about 577 feet, shadow 1,000 feet, sun angle 30 degrees

Illustration only. The upright line is the mountain's height. The ray is the sloping side. This is not a surveyed peak.

Tangent still means height divided by base.

Height = shadow length × tan(Sun angle).

Suppose a shadow is 1,000 feet long and the Sun is 30 degrees up. Tangent says the height is about 0.577 times the shadow length.

This is an invented example, not Galileo's data.

1,000 feet × 0.577 ≈ 577 feet.

This simple method also works on the Moon, approximately. You need the true ground length of the shadow, not an uncorrected length in a photograph, and the Sun's angle at that lunar place. The assumptions are level ground and sunlight traveling in the same direction. Real measurements may need corrections for slopes, the Moon's curvature, and the viewing angle. Mary McIntyre's account in BBC Sky at Night Magazine, "How to measure craters on the Moon," is the place that states the shadow method and those limits.

Galileo's Moon: what he actually did

The light reached the peaks first. In 1610 Galileo described bright mountaintops shining inside the Moon's dark region. Picture sunrise on Earth: a high peak can catch the Sun while the valley below is still dark.

He measured how far those bright peaks lay beyond the terminator, the boundary between lunar day and night. He compared that distance with the Moon's diameter, its full width through the center.

His published calculation used the Moon's curved shape and a right triangle. He used Pythagoras's rule, the relation between a right triangle's sides already used for the 3-4-5 triangle on the mathematics walk, to work out how far a peak had to rise to reach the sunlight.

That is not the shadow-times-tangent calculation above. Both turn light and triangles into a height. They are different methods.

Galileo reported a height "more than 4 Italian miles." Those are historical Italian miles, not automatically four of today's English miles. This was his estimate, not a modern survey. The English translation of *Sidereus Nuncius* by E. S. Carlos (1880), pages 28 to 30, is the text. Project Gutenberg ebook 46036.

Take this one idea with you. An angle fixes the triangle's proportions. One known length then tells the scale. Each function answers a different question about the same ladder. Sine: height compared with ladder length. Cosine: base compared with ladder length. Tangent: height compared with base. The names are new. The idea is comparing lengths.

The shadow example on this page is invented. It is not the number Galileo printed.

The same ladder sits inside the longer walk from algebra to calculus, Mathematics, one idea at a time.

Sources

- The lesson sheet itself, for the 20-foot ladder, the ratios 0.5, 0.866, and 0.577, and the invented 1,000-foot shadow.
- Galileo, *Sidereus Nuncius* (1610), E. S. Carlos translation (1880), pp. 28-30, Project Gutenberg ebook 46036.
- BBC Sky at Night Magazine, "How to measure craters on the Moon," on the shadow method and its limits.

Key facts

- Sine is height divided by the ladder. At 30 degrees that ratio is 1/2, so a 20-foot ladder reaches 10 feet.
- Cosine is base divided by the ladder. At 30 degrees the base is about 17.32 feet.
- Tangent is height divided by the base. At 30 degrees it is about 0.577. It is not the tangent line of a graph.
- Galileo's peak height used the Moon's curve and Pythagoras. It is not the shadow-times-tangent example, and "more than 4 Italian miles" is not four English miles.

The sheet is One ladder, three questions (PDF).